



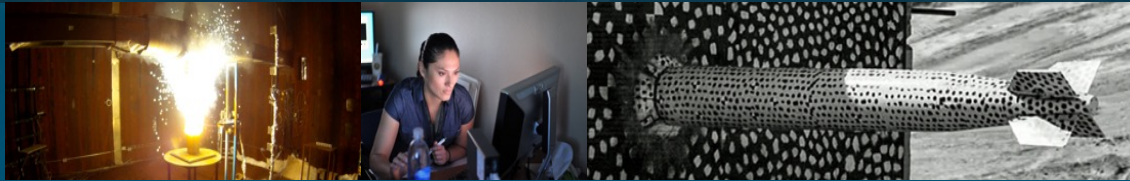
NLIT Summit 2019

National Laboratories Information Technology

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Viz Wars: Tableau VS Shiny



PRESENTED BY

Patrick Carlson & Adrian Perez



Sandia National Laboratories is a multimission laboratory managed and operated by National Technology & Engineering Solutions of Sandia, LLC, a wholly owned subsidiary of Honeywell International Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525.



We will discuss our experiences, where each tool excels, and their pros and cons. Both tools serve a particular niche.



Patrick Carlson, PhD

Education:

- PhD in Human-Computer Interaction

Sandia National Labs

- Data Scientist (April 2017)
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Adrian Perez

Education

- MS in Applied Statistics

Sandia National Labs

- Data Scientist (July 2018)
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Department Visualization Experience



50+ Tableau visualization created

20+ Shiny visualization created



Data Sciences Department

What we do

Sandia National Laboratories Data Science Department



Data Visualizations

- Tableau, Shiny, SSRS

Data Virtualization

- TIBCO

Data Transformations

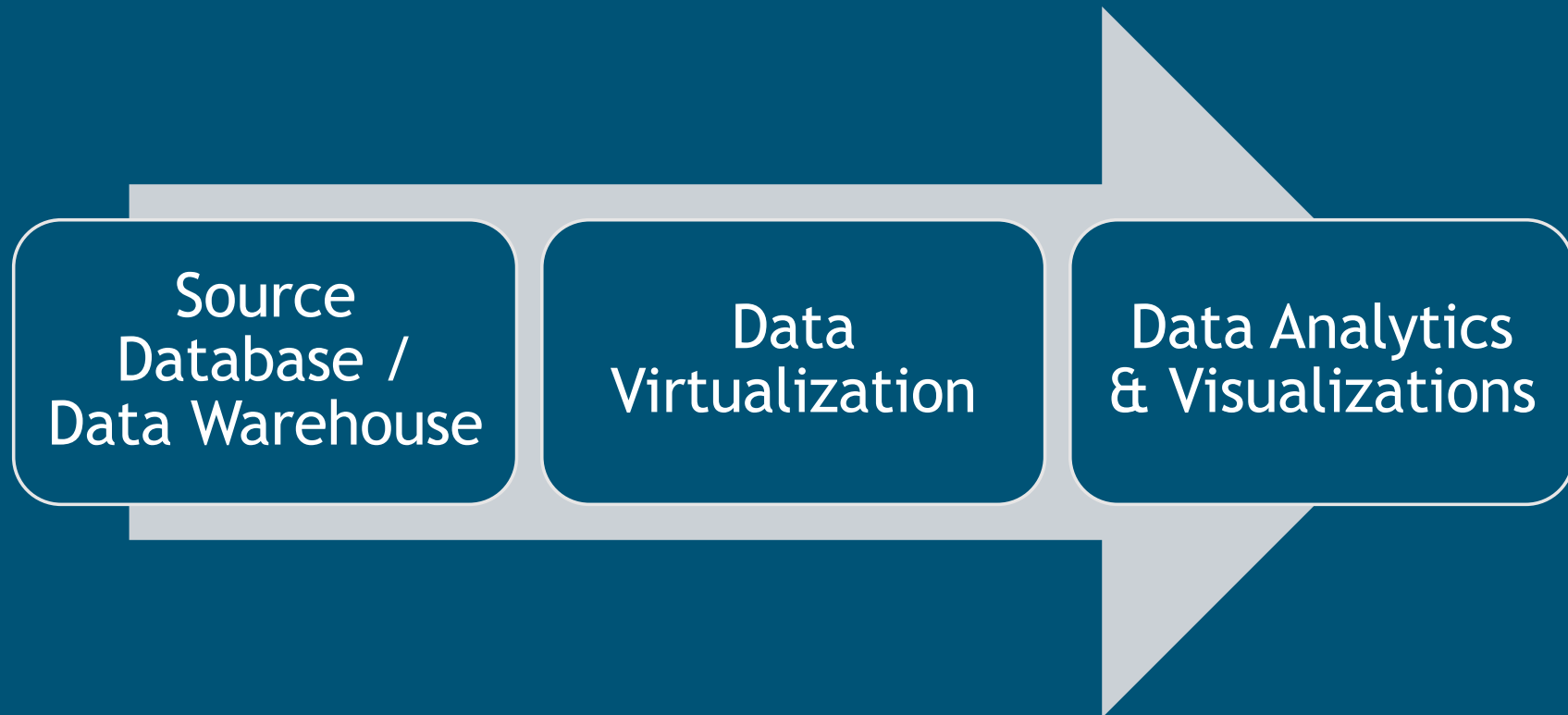
- ODI, SSIS

Application Development

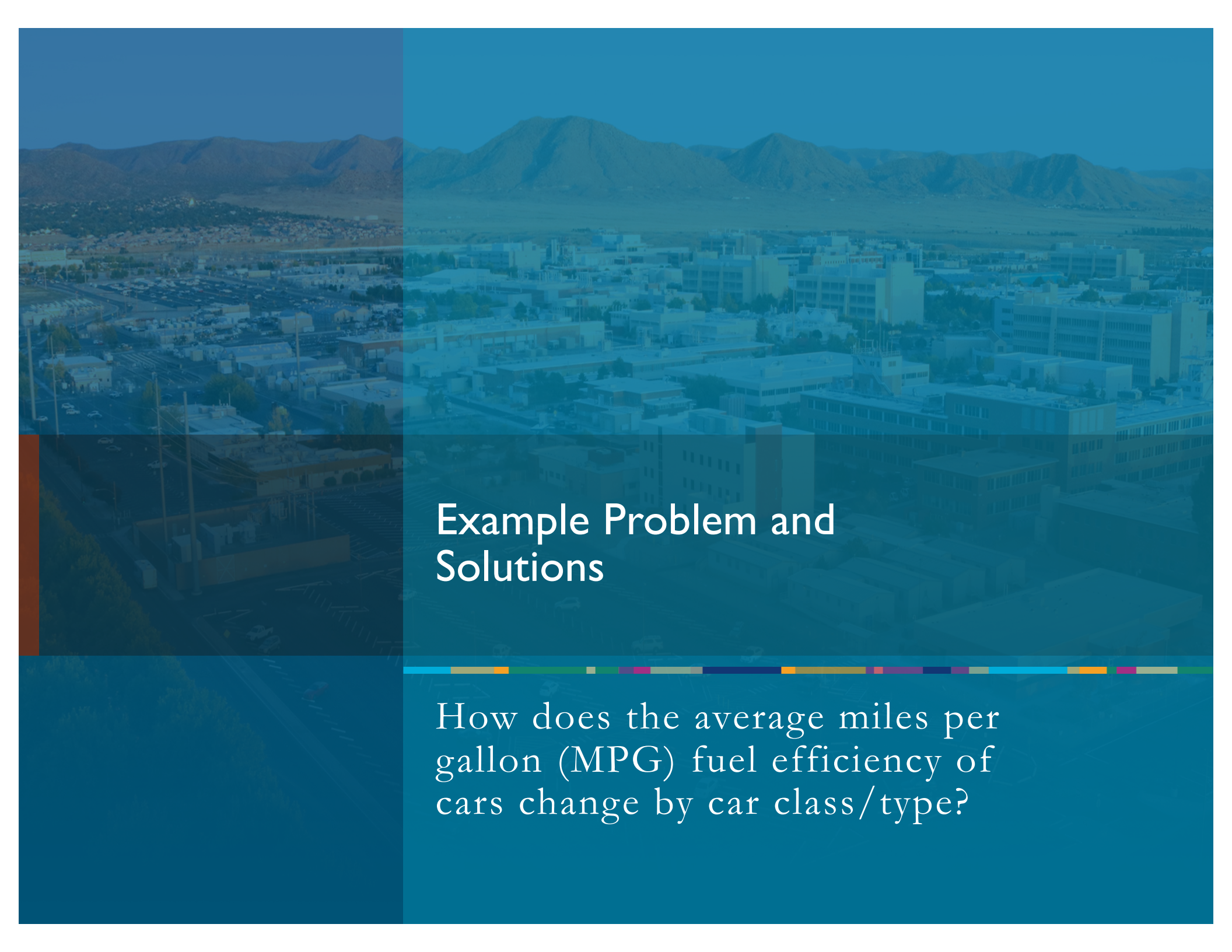
- Web Design, JavaScript, Java Spring Boot, Angular, Docker

Analytics and Machine Learning

- R, Python, Microsoft Machine Learning Server







Example Problem and Solutions

How does the average miles per gallon (MPG) fuel efficiency of cars change by car class/type?



Source: R's ggplot2 mpg

Columns

- Manufacturer
- Engine displacement
- Year of manufacture
- Number of cylinders
- Type of transmission
- Front / rear / 4WD
- City miles per gallon
- Highway miles per gallon
- Fuel type
- Type of car

Quick Facts: Tableau



Desktop client for authoring

Tableau server for hosting and sharing

Drag-and-drop interface for authoring



File Data Server Help

Connect

To a File

- Microsoft Excel
- Text file
- JSON file
- Microsoft Access
- PDF file
- Spatial file
- Statistical file
- More...

To a Server

- Tableau Server
- Microsoft SQL Server
- MySQL
- Oracle
- Amazon Redshift
- More...

Saved Data Sources

- job_structure_history ...
- My Superstore
- Sample - Coffee Chain...
- Sample - EU Superstore
- Sample - Superstore - ...
- Sample - Superstore S...

Open

Corporate Weekly ...

Employee Educatio...

MPG_Tableau_Exa...

Open a Workbook

Discover

Training

- Getting Started
- Connecting to Data
- Visual Analytics
- Understanding Tableau
- More training videos...

Resources

- Get Tableau Prep
- Blog - Tableau Conference Europe customer speaker applications are now open
- Forums

Sample Workbooks

Superstore

Regional

World Indicators

More Samples

VIZ OF THE WEEK

Are states financially healthy? →



The screenshot displays the Tableau Desktop interface. The top navigation bar includes icons for navigation, data source management, and view options. The left sidebar contains the 'Data' pane with a dropdown menu set to 'mpg'. Below this, the 'Dimensions' list includes 'Class', 'Drv', 'Fl', 'Manufacturer', 'Model', 'Trans', 'Year', and 'Measure Names'. The 'Measures' list includes 'Cty', 'Cyl', 'Displ', 'Hwy', 'Number of Records', and 'Measure Values'. The central workspace is divided into four shelves: 'Columns', 'Rows', 'Filters', and 'Marks'. The 'Marks' shelf is currently set to 'Automatic' and includes options for 'Color', 'Size', 'Text', 'Detail', and 'Tooltip'. The main visualization area is labeled 'Sheet 1' and contains the text 'Drop field here' in the top right and 'Drop field here' in the bottom left. The bottom status bar shows 'Data Source' and 'Sheet 1'.



The screenshot displays the Tableau desktop application interface. The top navigation bar includes various icons for navigation and analysis, a 'Standard' view selector, and a 'Show Me' button. The main workspace is divided into several panes:

- Data**: Shows the 'mpg' data source.
- Dimensions**: Lists fields such as Class, Drv, Fl, Manufacturer, Model, Trans, Year, and Measure Names.
- Measures**: Lists fields such as Cty, Cyl, Displ, Hwy, MPG, Number of Records, and Measure Values.
- Marks**: Contains a dropdown menu set to 'Automatic' and buttons for Color, Size, Text, Detail, and Tooltip.
- Columns** and **Rows**: Empty shelves for placing dimension fields.
- Filters**: An empty shelf for placing dimension fields.
- Sheet 1**: The main visualization area, currently blank, with 'Drop field here' prompts.

The bottom status bar shows 'Data Source', 'Sheet 1', and icons for saving and sharing.

Strengths: Tableau



Easy to learn and simple to use

Drag and drop interface

Fast visualizations on the fly

Customize visualizations (colors, fonts, etc.)

Industry leader

Lots of data connectors

Commercial support and updates

Documentation and example visualizations online

Permissions and administration through web-UI

Weaknesses: Tableau



Manual review of calculations and data pipeline

No templates in Tableau 10.4, start fresh each time

No debugging / visibility into calculations

Cost

Requires desktop license for authoring

Minimal version-control

Best Use Cases: Tableau



Quick graphing and fast turnaround

Ad-hoc work

Lives temporarily

Collaborative projects

Multiple visualizations from a single dataset

Enterprise delivery

Quick Facts: Shiny



RStudio desktop IDE for authoring and local testing

Shiny Server or Shiny Server Pro (paid) for hosting visualizations

Uses the R programming language

Can leverage CRAN packages

Can do more than just visualize data



File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins - Project: (None) -

app.R x

```

1 library(shiny)
2 library(dplyr) # for easy dataframe manipulation
3 library(readr)
4 library(ggplot2) # for the "mpg" dataset
5 library(plotly) # for the interactive graph
6 library(shinyWidgets) # for the multi-select dropdown year filter
7
8 # save dataset for use in Tableau
9 # https://ggplot2.tidyverse.org/reference/mpg.html
10 # write_csv(mpg, path="mpg.csv")
11
12 mpg <- read_csv("mpg.csv")
13

```

6:28 (Top Level) R Script

Environment History Connections

Import Dataset - List -

Global Environment -

Environment is empty

Files Plots Packages Help Viewer

New Folder Delete Rename More

2019 > Tableau_vs_Shiny > Shiny-MPG-example ...

	Name	Size
	..	
	.Rhistory	17.9
	app.R	1.7 K
	mpg.csv	12.9
	MPG_Tableau_Example.twbx	9.3 K

Console Terminal x

Y:/Conferences/NLIT_2019/Tableau_vs_Shiny/Shiny-MPG-example/ ↗

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Microsoft R Open 3.5.1
The enhanced R distribution from Microsoft
Microsoft packages Copyright (C) 2018 Microsoft Corporation

Using the Intel MKL for parallel mathematical computing (using 12 cores).

Default CRAN mirror snapshot taken on 2018-08-01.
See: <https://mran.microsoft.com/>.

```

> setwd("Y:/Conferences/NLIT_2019/Tableau_vs_Shiny/Shiny-MPG-example")
>

```



File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins - Project: (None)

app.R x

```

8 # save dataset for use in Tableau
9 # https://ggplot2.tidyverse.org/reference/mpg.html
10 # write_csv(mpg, path="mpg.csv")
11
12 mpg <- read_csv("mpg.csv")
13
14 # calculate average MPG by combining city and highway MPG (rough approximation)
15 mpg_df <- mpg %>% dplyr::mutate(avg_mpg = ((cty + hwy)/2))
16
17 ui <- fluidPage(
18   titlePanel("Shiny Average MPG Example").
19 )
20
54:43 server(input, output)
R Script

```

Environment History Connections

Global Environment -

Data

mpg 234 obs. of 11 var...

Files Plots Packages Help Viewer

New Folder Delete Rename More

2019 > Tableau_vs_Shiny > Shiny-MPG-example

	Name	Size
	..	
	.Rhistory	17.9
	app.R	1.8 K
	mpg.csv	12.9
	MPG_Tableau_Example.twbx	9.3 K

Console Terminal x

```

Y:/Conferences/NLIT_2019/Tableau_vs_Shiny/Shiny-MPG-example/
manufacturer = col_character(),
model = col_character(),
displ = col_double(),
year = col_double(),
cyl = col_double(),
trans = col_character(),
drv = col_character(),
cty = col_double(),
hwy = col_double(),
fl = col_character(),
class = col_character()
)

Listening on http://127.0.0.1:6852

>

```

MPG Calculation (approximation)



```
# calculate average MPG by combining city and highway MPG (rough approximation)  
mpg_df <- mpg %>% dplyr::mutate(avg_mpg = ((cty + hwy)/2))
```



```
ui <- fluidPage(  
  
  titlePanel("Shiny Average MPG Example"),  
  
  # Sidebar with a picker for year  
  sidebarLayout(  
    sidebarPanel(  
      pickerInput("year",  
                  label="Filter Year:",  
                  choices = sort(unique(mpg_df$year)),  
                  multiple = TRUE,  
                  options = list(`actions-box` = TRUE),  
                  selected = sort(unique(mpg_df$year)))  
    ),  
  
    mainPanel(  
      plotlyOutput("plotlyBargraph")  
    )  
  )  
)
```



```
server <- function(input, output) {

  getFilteredDF <- reactive({
    return(mpg_df %>% dplyr::filter(year %in% input$year))
  })

  getPlotlyDF <- reactive({
    return(getFilteredDF() %>% dplyr::group_by(class) %>%
      summarize(avg_mpg_class = mean(avg_mpg)) %>% as.data.frame()
    )
  })

  output$plotlyBargraph <- renderPlotly({
    # https://plot.ly/r/bar-charts/
    plot_ly(
      x = getPlotlyDF()$class,
      y = getPlotlyDF()$avg_mpg_class,
      name = "Average MPG by Car Type",
      type = "bar"
    ) %>%
    layout(
      xaxis = list(title="Car Type"),
      yaxis = list(title="Average MPG")
    )
  })
}
```

Strengths: Shiny



Connects to almost any data source

Provides nearly every chart and graph type

Repeatable / reproduceable

Powerful data manipulation and transformation

Highly customizable

Code can be version-controlled

Free and Open Source

Documentation and example material

Diversity of CRAN packages and leverage Open Source community

RMarkdown, common R language from Data Scientists / Analysts

Weaknesses: Shiny



Requires R programming language and technical expertise

Sometimes requires HTML / Javascript knowledge

Updating server R packages, keeping environments in sync

Less commercial support options

Less popular in industry

Future viability of R over Python or other languages

Best Use Cases: Shiny



High customization

Flexible UI

Data source types not supported by Tableau

Specialty graphs (Sankey, network diagram, etc.)

More data processing or transformations relative to Tableau

Helpful Shiny Packages



Tidyverse (dplyr, stringr, etc.)

Plotly (graphing)

DT (datatables.net)

networkD3 (graph network)

ShinyDashboard (consistent theme/layout)

shinyBS (Bootstrap)

ShinyCSSloaders (loading spinners)

Leaflet (mapping)

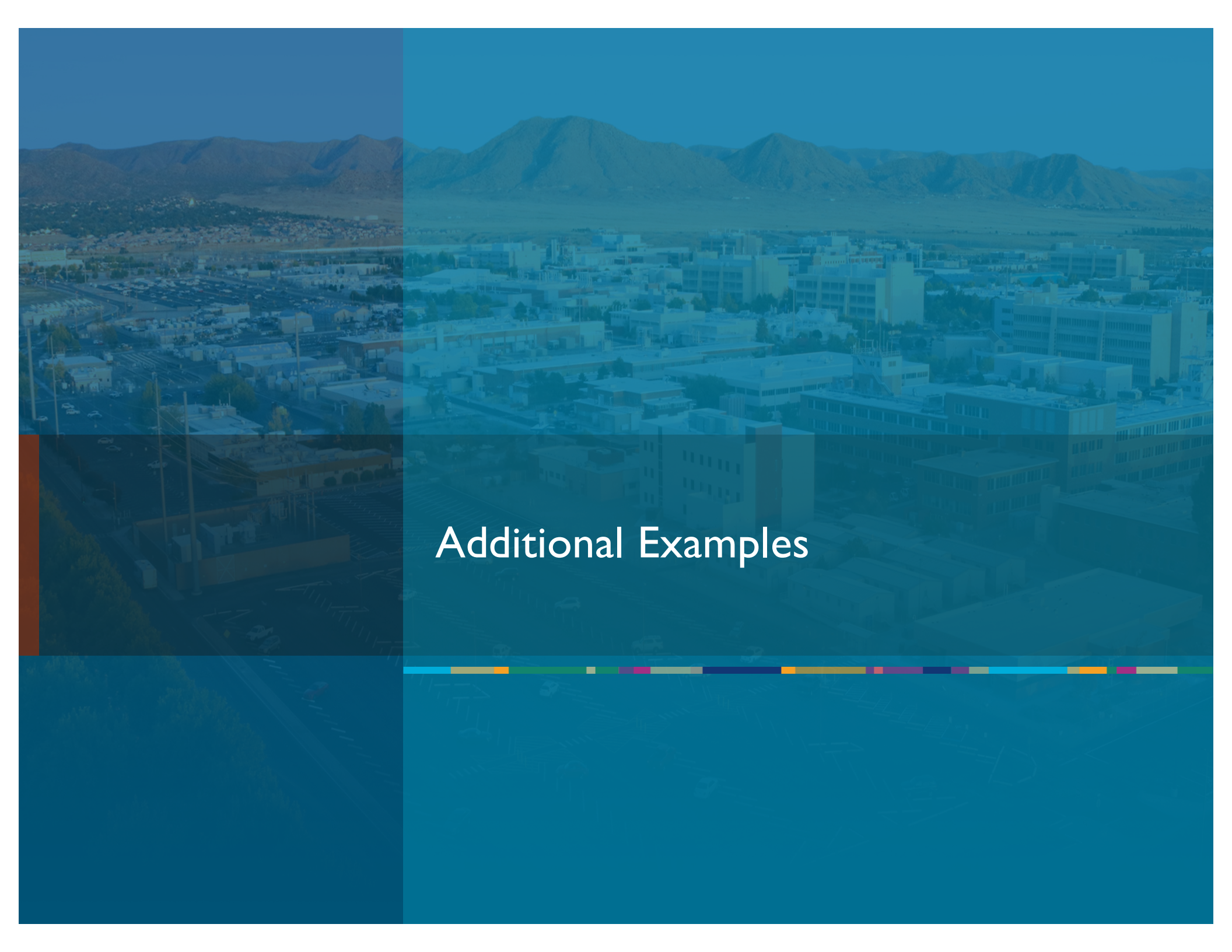
shinyWidgets (additional UI widgets)



Both tools serve particular niche areas

Consider:

- Technical skills
- Investment (cost, training)
- User base / size



Additional Examples

31 Tableau – Employee Education

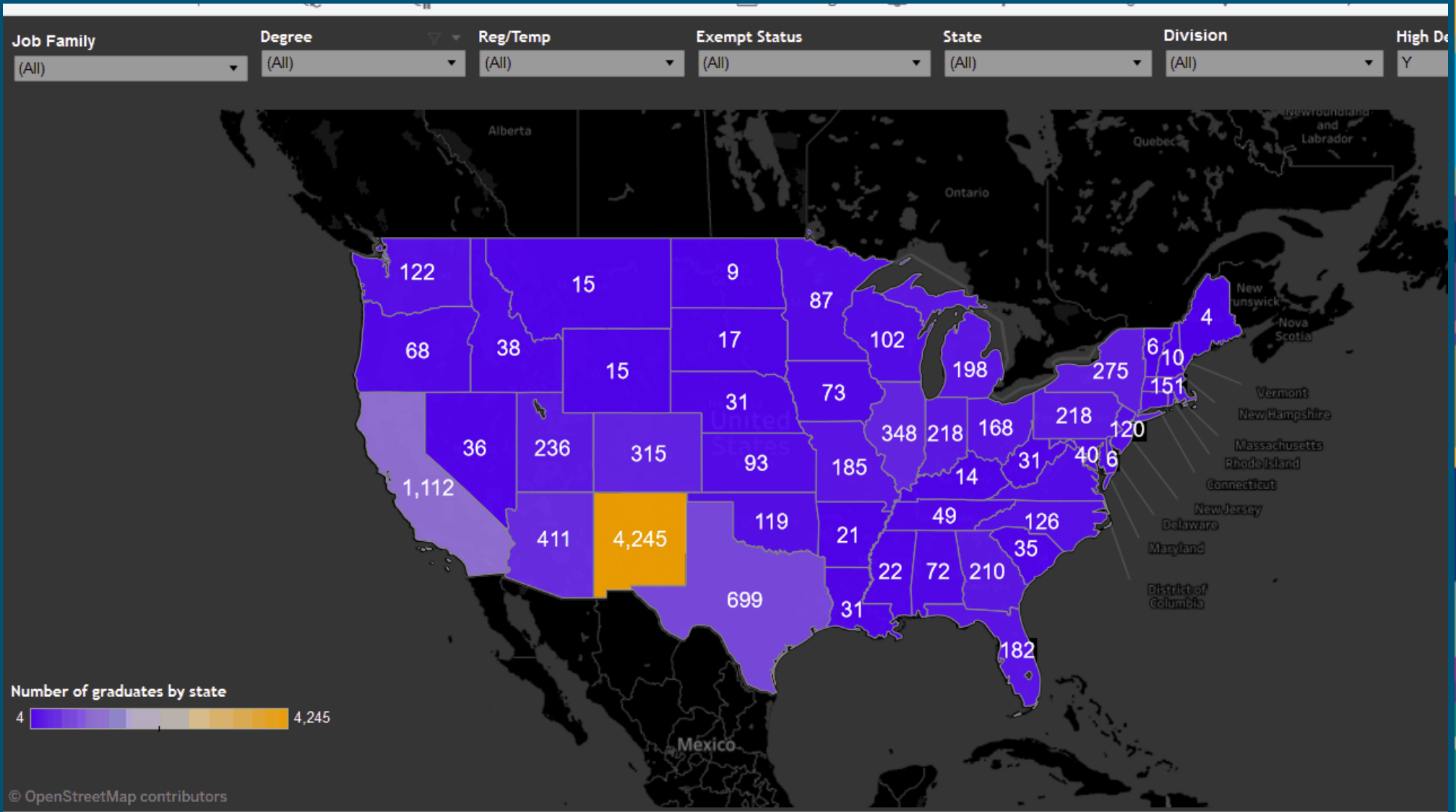
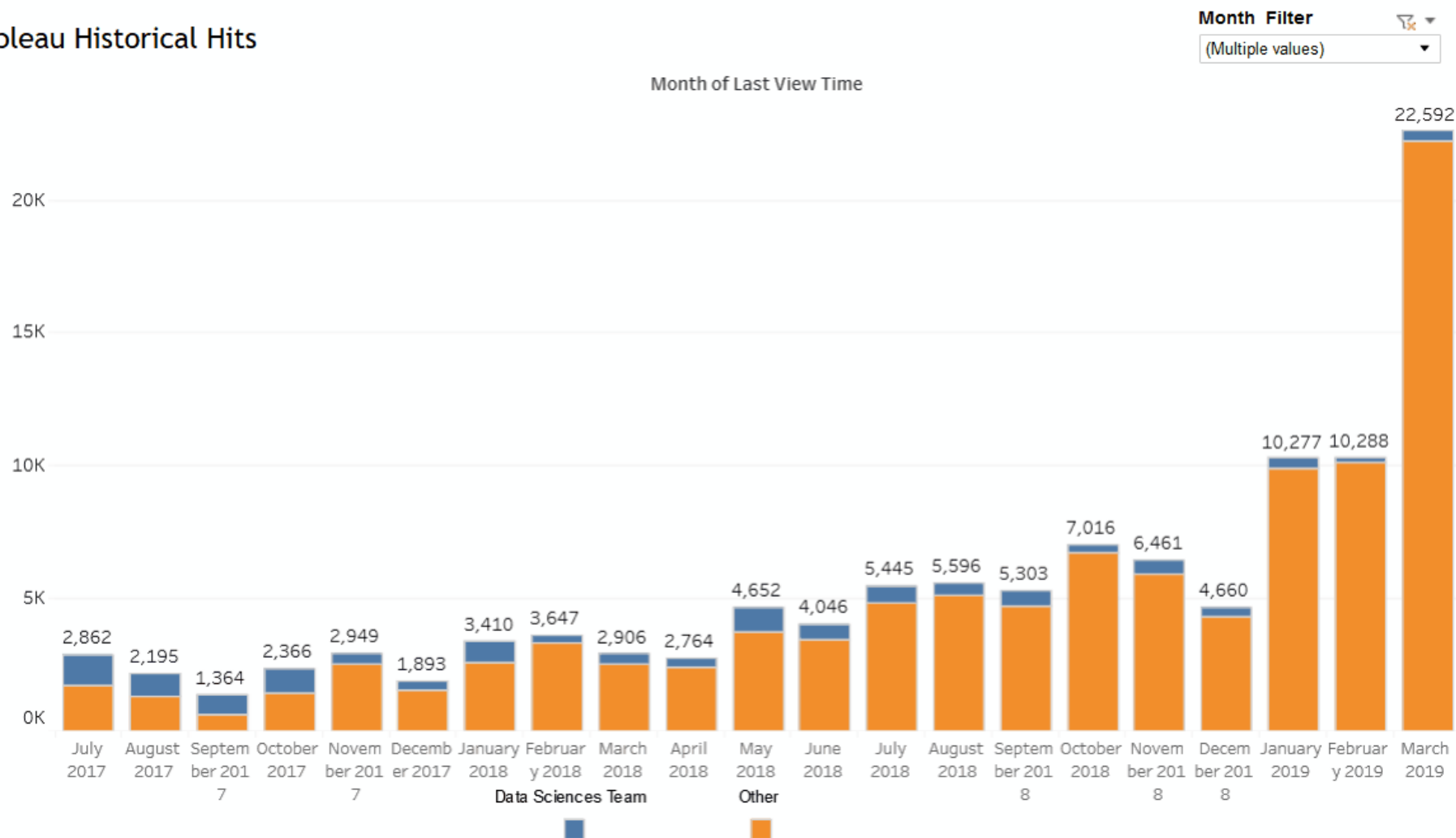




Tableau Historical Hits





Internal Movement - Sankey

Sankey Viz About

FY(s)

1998, 1999, 20 ▾

Starting
Division(s)

01000, 02000, ▾

Ending Division(s)

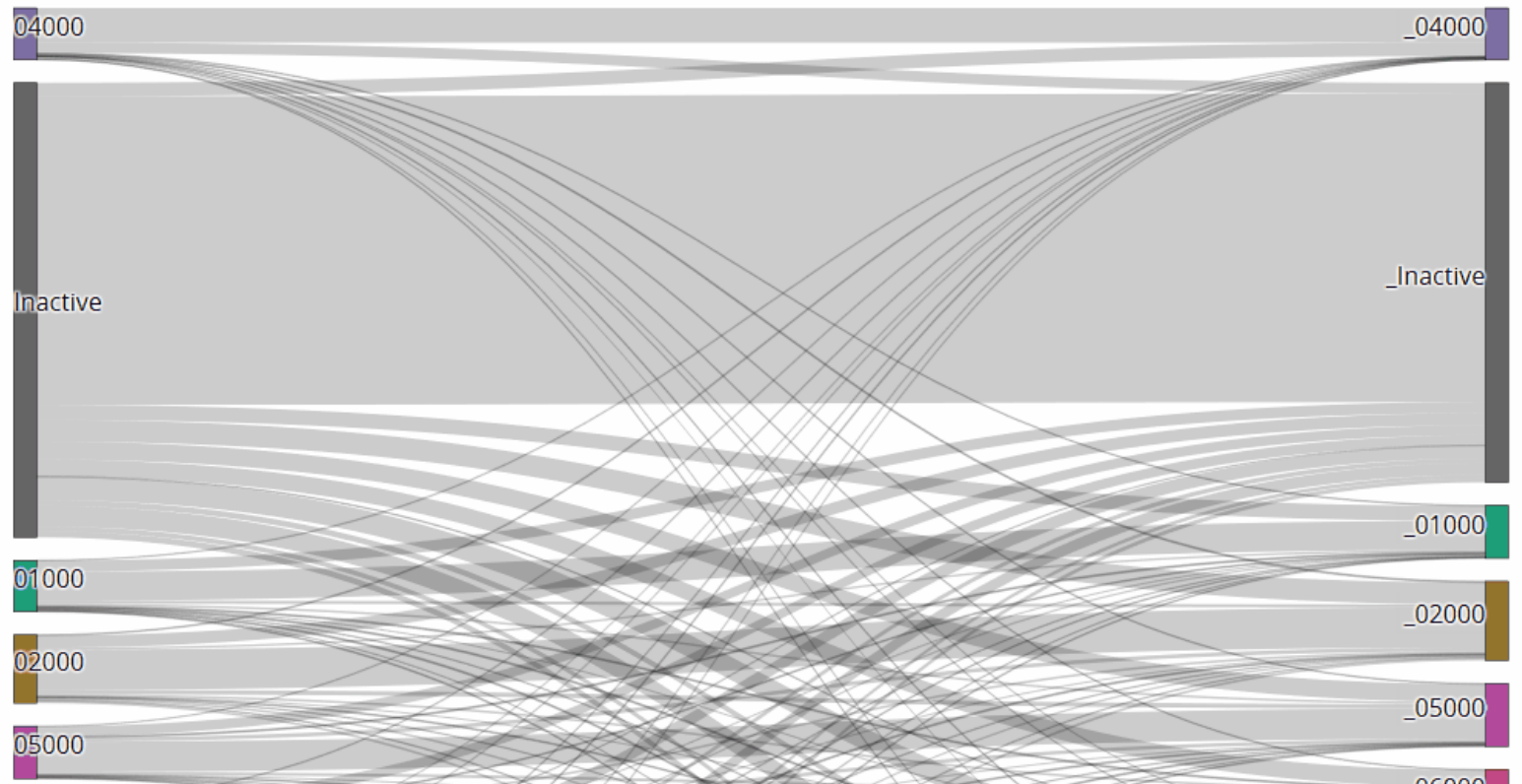
_01000, _02000 ▾

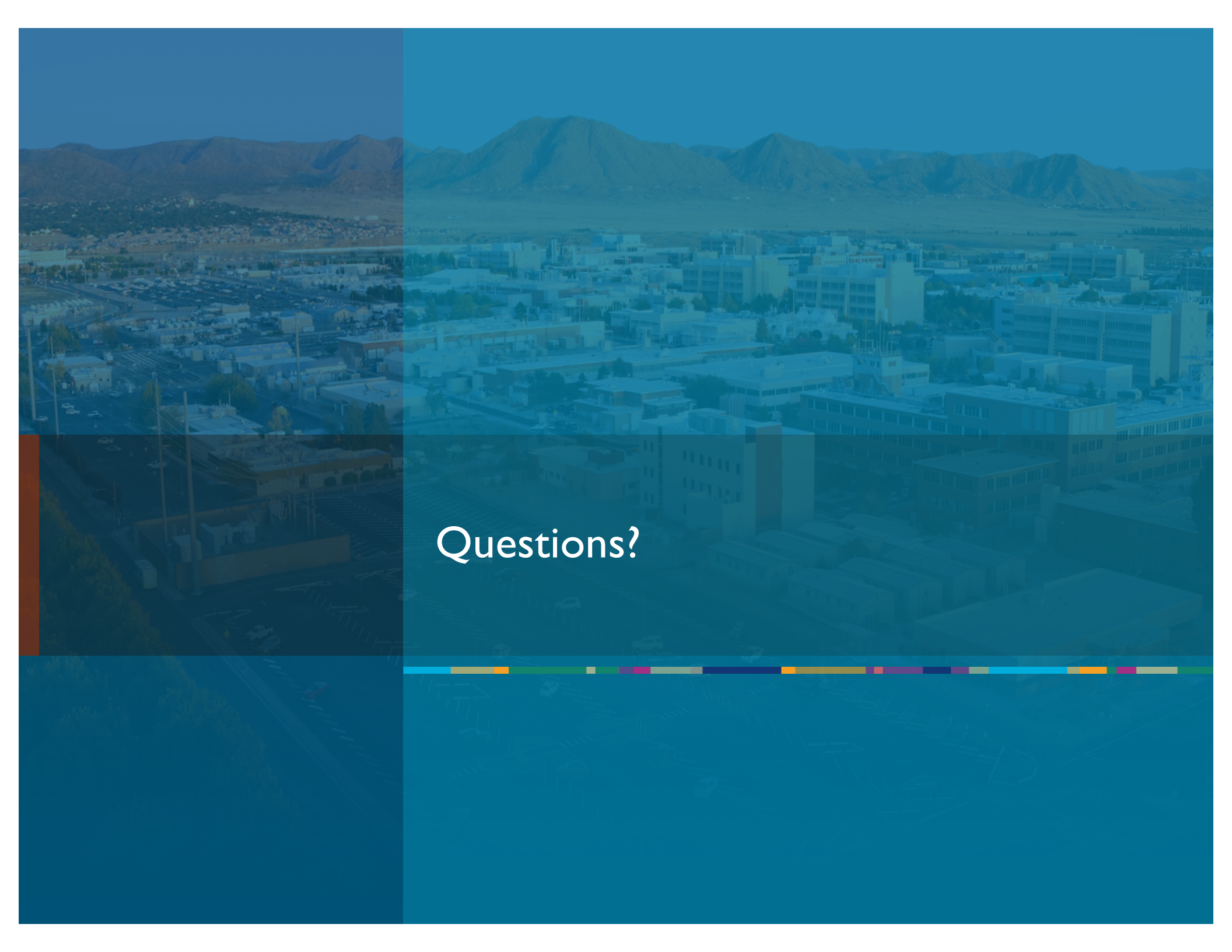
Sankey

Counts Table

People Table

Internal Movement





Questions?

“MPG” Example

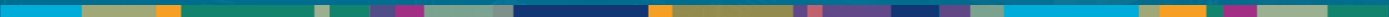


Tableau and Shiny example files and R code:

<https://github.com/carlsonp/NLIT-presentation-2019>



Appendix





Tableau

- <https://www.tableau.com/learn/training>
- <https://public.tableau.com/en-us/s/authors/>
- <https://public.tableau.com/en-us/s/gallery>

Shiny

- <https://shiny.rstudio.com/tutorial/>
- [R for Data Science](#) by Garrett Grolemund & Hadley Wickham
- [Interactive web-based data visualization with R, plotly, and shiny](#) by Rob Kabacoff
- [R Graphics Cookbook](#) by Winston Chang